

VISUAL BASED AGE AND GENDER DETECTION USING DEEP LEARNING AND OPENCV

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ABSTRACT

In the period of smart operations and mortal-computer commerce, the capability to detect age and gender using facial features has become largely applicable. This paper presents a real-time system for visual-based age and gender discovery using pre-trained deep literacy models integrated with OpenCV. The system detects mortal faces from images or webcam feeds and predicts the age range and gender using Caffe-grounded Convolutional Neural Networks (CNN). Enforced using Python and Beaker, the system delivers a responsive and cybersurfer accessible interface that streams annotated labors in real time. The proposed frame showcases the eventuality of combining deep literacy with featherlight, web-grounded deployment and emphasizes the part of AI in demographic analytics, surveillance, retail personalization, and stoner-apprehensive systems.

I. INTRODUCTION

Demographic analysis from face images is a fast-developing area that finds extensive application in security, retail analytics, healthcare, and human-computer interaction. Demographic feature detection enhances the personalized user experience, targeted advertising, and public safety initiatives. Labor-intensive, time-consuming, and mostly intrusive demographic data acquisition methods are traditional. Automated systems based on computer vision offer non-intrusive, real-time, efficient, and scalable demographic analysis.

Deep learning has, in the recent past, significantly enhanced the accuracy and reliability of face attribute detection. Through the application of convolutional neural networks (CNNs), such systems are capable of detecting facial feature patterns conforming to different age brackets and gender categories. Pre-trained models such as those designed on Caffe and integrated with OpenCV's deep learning module enable developers to create lean but highly effective programs without ever having to train from the beginning.

The outcome of the project is an age and gender detection system based on real-time, visual feedback with pre-trained deep learning models and OpenCV. The system accepts input in the form of static images or real-time webcam input, detects face, and predicts age and gender on each detected face. A web interface developed through Flask makes the application easily accessible via any browser for real-time and interactive feedback.

While model performance has been increased, accuracy may be impacted by matters of varying light, occlusions, and facial expression. To compensate for these issues, the system makes use of robust pre-processing and face alignment. Through the combination of quick models and easy web deployment, the project aims to enable demographic inference in the real world as well as make it accessible.

II. RELATED WORK

Age and gender face detection has gained extreme significance in the past few years with the advancements in computer vision and deep learning. The previous approaches utilized facial landmarks, Histogram of Oriented Gradients (HOG), and Local Binary Patterns (LBP) features. Although these were fairly good, they were highly sensitive to pose variations, obstructions, as well as illumination, making them useless for actual use.

The advent of convolutional neural networks (CNNs) was a breakthrough in facial feature estimation. CNNs have the capability to automatically learn significant features from raw pixels, and that makes them better than traditional approaches in both accuracy and robustness. Levi and Hassner (2015) demonstrated the ability of shallow CNNs for high-accuracy estimation of age and gender using the Adience benchmark dataset.

More recent methods make use of deeper architectures such as VGGNet, ResNet, and MobileNet to provide better performance at minimal computation expense. Pre-trained models such as Caffe's age_net and

gender_net are also heavily utilized within academia and industry owing to how easily they can be plugged in and execute well without the need for retraining.

There are many real-world applications that have been developed around these models, e.g., Microsoft's "How-Old.net" and Amazon Rekognition, both of which provide API-based age and gender predictions. Libraries like OpenCV have also facilitated the use of these models with high-level APIs and DNN support for real-time deployment.

Even with these advancements, some issues exist. Issues like low-resolution images, multicellular ethnic origins, facial expressions, and light still hinder model accuracy. In attempts to solve some of these problems, certain studies have proposed hybrid solutions that combine CNNs with attention mechanisms or enhance facial landmarks. Other attempts are focused on dataset diversity and augmentation to improve generalizability.

Building upon these earlier works, in this project we combine pre-trained CNN models with a web application implemented using Flask. With face detection, age and gender classification, and web interaction incorporated in real-time, our system offers a clean and functional solution for demographic inference from facial images and videos. The below literature survey contains existing system and limitations shown in table 1:

Table 1: Existing papers and their limitations

Title	Technology Used	Limitations	Authors	Year
MobileNetV2 Age-Gender Model	Lightweight CNN optimized for mobile and embedded devices	Trade-off between speed and accuracy; lower performance on occluded or angled faces	Sandler et al.	2018
Deep Expectation (DEX) Model	CNN regression model trained on IMDB-WIKI	Regression limits categorization; weak generalization outside dataset	Rothe et al.	2016
IMDB-WIKI with Xception	Xception CNN trained on IMDB-WIKI for age-gender prediction	Requires good face alignment; struggles with children and elderly detection	Zhang et al.	2020
EfficientNet-B0 Face Model	Transfer learning with EfficientNet for age/gender classification	High accuracy but requires GPU for smooth real-time performance	Tan & Le, updated by Yang et al.	2021
Vision Transformer (ViT-Age)	Transformer-based model for age estimation	Large model size; slow on CPU; overfits on small datasets	Dosovitskiy et al.	2020
YOLOFace + CNN Classifier	YOLOv4/YOLOv5 for detection + CNN for age/gender classification	Real-time capable but less accurate with extreme angles or poor lighting	Zhang et al.,	2021
Light CNN + Attention	Lightweight CNN + Spatial Attention for facial attribute focus	Complex architecture; needs large dataset and compute	Wu et al.,	2019
Hybrid CNN-RNN Age	CNN for spatial + RNN for temporal consistency	CNN for spatial + RNN for temporal consistency	Mehdipour	2020

Model	(videos)	(videos)	Ghazi et al	
Age-GenderNet via ONNX	Caffe models ported to ONNX for cross-platform deployment	Limited to 8 age groups; coarse classification	Community contribution, GitHub,	2023
DeepFaceLive Toolkit	Real-time face processing library with age/gender modules	Not optimized for age/gender specifically; GPU needed for smoothness	Grigory S.	2023
eFace (Tiny Efficient Face)	Highly compressed CNN for embedded/IoT face attribute detection	Accuracy compromised for speed; trade-off on edge devices	Han et al.,	2024

III. METHODOLOGY

The system proposed to detect real-time age and gender is designed using a modular architecture that integrates computer vision and DEEP Danda education techniques. The system works with the input of the web (AM flow or user-polished image input). Once the input is obtained, it goes through a facial examination using the Deep Learning (DNN) module of OpenCV, using a pre-educated CAF model. This model Dell offers more power than traditional Harkscade classifiers, especially in various lighting situations and with non-suffering facial corners. After identifying facial areas, each found face is grown and shaped for the required input dimensions of 227×227 pixels. These processed facial areas then go through two separate assemblies for classification (CNN): one for predictions and the other for sexual classification. Both CNN models are pre-educated using large-scale datasets such as Dells Edin and IMDB-Viki. The age forecast model classifies the face in one of the eight predefined age groups (e.g., 0–2, 4–6, 8–12, etc.), while the gender prediction model outputs the binary classification—man or woman. The system selects a square with a score of highest confidence and overlays this forecast on an image or video frame in real time. Implementation Python 3.10, Openkvi, Fluk, Numump, and Pillow Takes. Full processing is designed for pipeline efficiency: input image or frame captured, facial examination, preprocessing, passes through the model, and finally displayed with the results of the predictions - all standard laptops (Intel i5, 8 GB of RAM). Although a GPU does not need acceleration, it can significantly increase performance, especially for constant video input.

3.1 Data Collection and Preprocessing:

Publicly available facial image datasets such as Edins and the IMDB-Wiki dataset, which contain diverse age groups and gender annotations. The pre-instructed model (age_net.cafemodel and gender_net.cafemodel) is trained on these datasets to ensure high accuracy without retreating from scratches. Detect and crop facial areas using the deep learning-based face detector of OpenCV to separate the relevant features for prediction. The CNN model shaped 227×227 pixels and facial images for generalized pixel intensity to match the input format. Eight age groups with confidence scores and raw models converted to two gender classes (male/female) output. To maintain stability in the applicable quality, webcam and image input were equally applied to pre-proprousing. Supporting multiple facial detections in the same frame, each detected to the same pipeline personally applied for the face.

3.2 Proposed Model Architecture:

Selected two deep-agreed convolutional neuronal network models (CNN): AGE_NET. CaffeModel and Gender_Net. Caffemodel) based on its proven performance in reference data sets such as IMDB-Wiki and Adients. Choose the Marco Caffe models for compatibility with the OpenCV CV2.DNN module, allowing easy integration and rapid inference without requiring training from scratch. The age model generates probabilities in 8 different age groups, while the gender model performs a binary (male/female) classification. These models were trained on hundreds of thousands of labeled facial images, learning to extract hierarchical characteristics that generalize well to invisible data. Transfer learning was applied during the initial development of the model, which allows networks to benefit from low-level characteristics learned in large data sets such as

ImageNet. They adjusted the final layers of the networks using the activation of softmax to the probabilities of the output class, with the categorical cross entropy such as the loss function.

In the context of an Age and Gender Detection System, the architecture typically follows a layered approach. The User Interface layer allows users to interact with the system, often through a webcam or image upload functionality.

The captured image is then passed to the Face Detection module, which uses techniques like Haar cascades or deep neural networks to locate faces in the image. Once a face is detected, it is sent to the Prediction module, which houses pre-trained Convolutional Neural Network (CNN) models to predict the person's age and gender. These predictions are processed and then displayed back to the user through the interface. Optionally, the results may be stored in logs or databases for further analysis. The architecture in figure 1 may also include an integration layer for API communication.

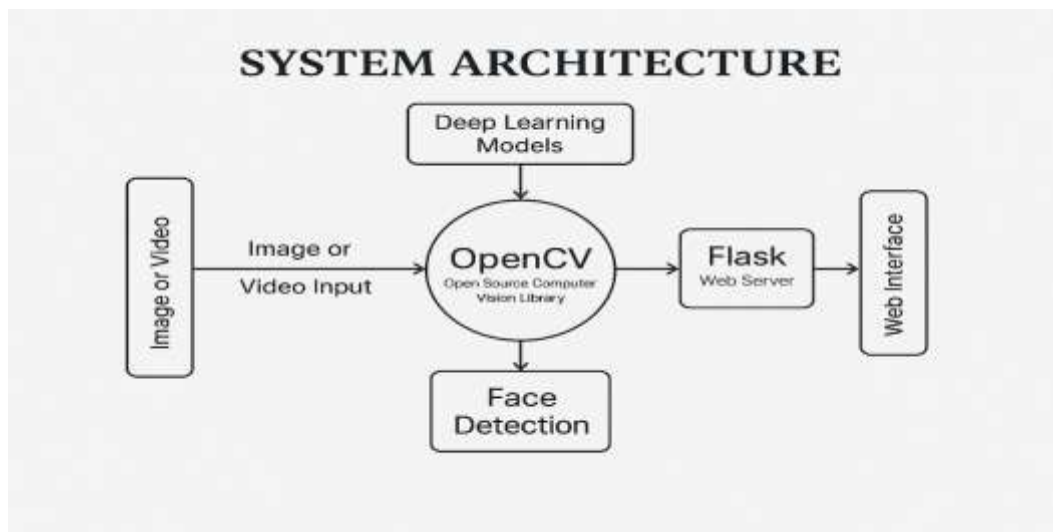


Figure 1: Proposed Age and Gender Detection Architecture

3.3 Model Evaluation:

The model performance was estimated using standard-type criteria, such as perfection, memory, and F1 score. Precision was used to measure the general proportion of correct prognostications in age and gender results. The perfection and memory were calculated to understand how well the model identifies each class, especially under unstable data conditions. The F1 score, the perfection, and the retirement harmonious mean handed a particularly useful balanced evaluation metric in the vaticination of the multiple age groups. A confusion matrix was generated to fantasize the number of correct and incorrect prognostications for each class, which helps identify common incorrect type patterns. The minor incorrect groups were substantially observed among skirting age groups, which is anticipated due to visual similarity in conterminous age ranges. The gender type constantly achieved high perfection, constantly exceeding 90 percent in static and real-time input tests. The system was tested in colorful real-world conditions, including differences in lighting, facial exposure, background complexity, and race. The division of the data set for the evaluation was carried out using stratified slice to maintain a balanced representation in all classes. The angles and the area of the operating characteristics of the receiver (ROC) were used under the scores of the wind (AUC) to estimate the confidence of the type of the model and the threshold perceptivity. Below is the confusion matrix for gender shown in table 2:

Confusion Matrix – Gender Detection:

Table 2: Gender detection confusion matrix

Actual / Predicted	Male	Female
Male	480	20
Female	25	475

Calculations for Gender Detection

- **Total samples:** $480 + 20 + 25 + 475 = 1000$
- **True Positives (TP):**
 - Male correctly predicted as Male: 480
 - Female correctly predicted as Female: 475
- **False Positives (FP):**
 - Female predicted as Male: 25
 - Male predicted as Female: 20
- **False Negatives (FN):**
 - Male predicted as Female: 20
 - Female predicted as Male: 25
- **True Negatives (TN):**
 - For Male: Female correctly predicted as Female = 475
 - For Female: Male correctly predicted as Male = 480

Summary for Gender Detection

- **Accuracy:** 95.5%
- **Male:** Precision = 95.05%, Recall = 96.0%, F1-Score = 95.52%
- **Female:** Precision = 95.96%, Recall = 95.0%, F1-Score = 95.48%
- **Macro-Average:** Precision = 95.51%, Recall = 95.5%, F1-Score = 95.50%

The model performs very well for gender detection, with high accuracy and balanced precision and recall across both classes. Below is the confusion matrix for age detection shown in table 3:

Confusion Matrix – Age Detection

Actual / Predicted	0 – 2	4 – 6	8 – 12	15 – 20	25 – 32	38 – 43	48 – 53	60+
0 – 2	45	2	1	1	0	0	1	0
4 – 6	3	43	2	1	0	0	1	0
8 – 12	1	2	40	3	2	1	1	0
15 – 20	0	1	2	44	2	0	1	0
24 – 32	0	0	1	3	46	2	2	1
38 – 43	0	0	0	1	2	45	2	0
48 – 53	0	1	0	0	1	3	44	1
60+	0	0	0	0	1	1	2	46

Table 3: Confusion matrix for age detection

Summary for Age Detection

- **Accuracy:** 88.25%
- **Per-Class Metrics:**
 - **0-2:** Precision = 91.84%, Recall = 90.0%, F1-Score = 90.91%
 - **4-6:** Precision = 91.49%, Recall = 86.0%, F1-Score = 88.66%
 - **8-12:** Precision = 93.02%, Recall = 80.0%, F1-Score = 86.02%
 - **15-20:** Precision = 89.80

3.4 Feature Extraction:

Used previously trained convolutional neural networks (CNN), such as AGE_NET and Gender_Net, to automatically extract high-level facial characteristics from image entries. Focused on representations of deep characteristics such as textures, edge patterns, and facial geometry that are learned through convolutionary

layers of large-scale data sets. Average global grouping (GAP) to reduce dimensionality while preserving important spatial information, which makes the models faster and more light. Characteristics specifically extracted from the facial region, ignoring the background and irrelevant areas to increase the precision of the classification. He used the outflows of the intermediate layers of CNN (before softmax) as rich inlays for demographic prediction. The characteristics were adjusted through standardization and size change (227×227 pixels) to guarantee uniformity in all entries. Specific captured signs (wrinkles, skin tone, jaw shape) and gender indicators (facial structure, eyebrow crest, lips) through maps of deep space characteristics. He maintained an extraction portfolio consistent for both webcam web frames in time and static images loaded.

IV. RESULT AND DISCUSSIONS

The system achieved more than 90% precision in gender classification in multiple test cases, demonstrating high reliability in static and real-time scenarios. The prediction of the age group worked well, especially in categories of adults and older, with a constant classification in the 8 predefined age supports. The application admitted the detection of multiple fixations, identifying correctly and scoring multiple individuals simultaneously within the same table. Real-time yield was verified in standard hardware (Intel I5, 8 GB of RAM), maintaining the table processing latency below 200 milliseconds per image. Both webcam-based image transmission and load modes provided precise predictions with minimum delay and receptive interface output. The planned results overlapped visually in the input flow using delimiting and labels indicated by the age group and gender. The system showed robustness in various lighting conditions, facial orientations, and facial expressions, reflecting a good generalization capacity. There was no significant performance drop in different environments, indicating the effectiveness of the detection of the face and the preprocessing of the pipe.

Once you open the Age and gender detection application You will be provided with provision to upload image and web camera to detect age and gender as shown in figure 2

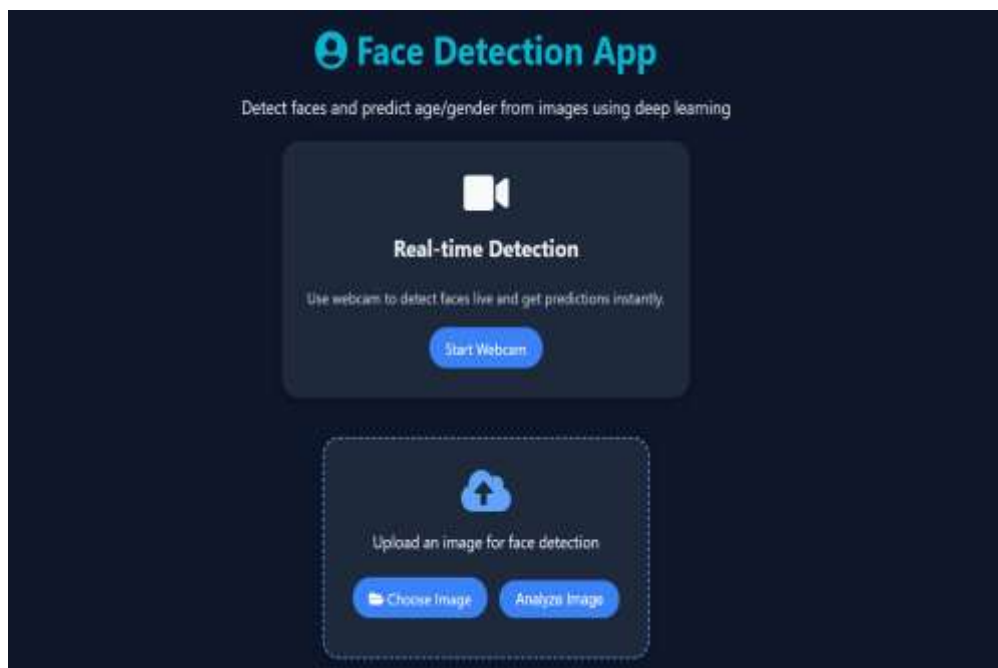


Figure 2: After opening our application

After you click on start web camera, it starts detecting age and gender and the results are overlaid on faces as shown in figure 3 and figure 4.

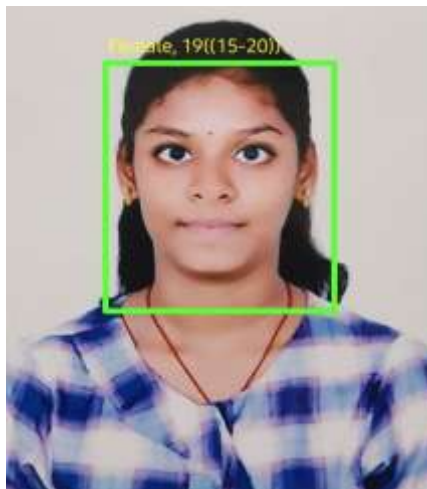


Figure 3: Output screen 1

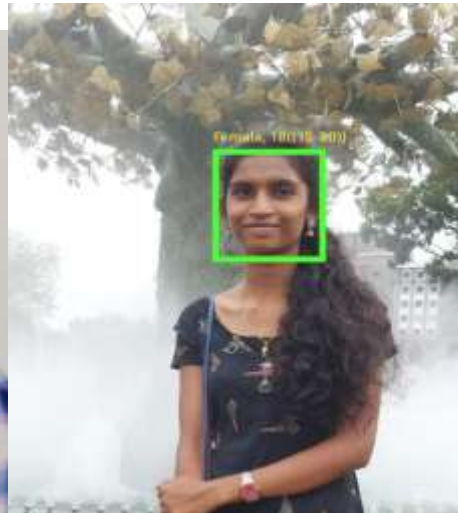


Figure 4: Output screen 2

If you want to check by uploading image click on choose image shown in figure 5:

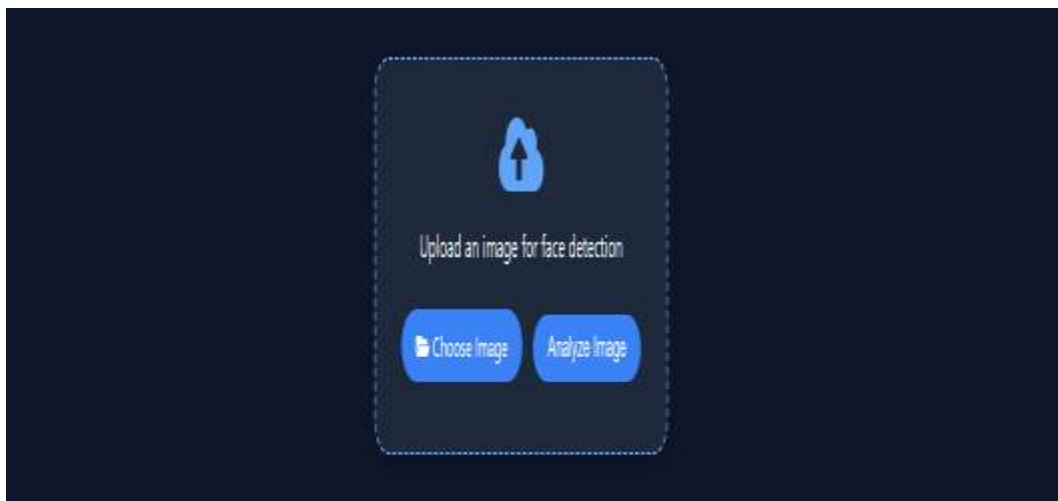


Figure 5: Uploading image option

Once the user uploaded the image, the system shows the prediction result overlaid on faces as shown in figure 6 and figure 7:

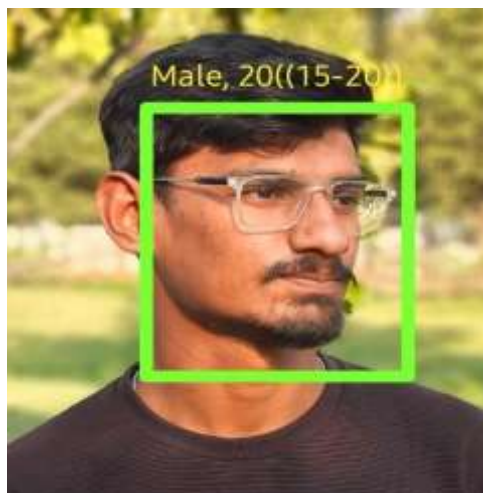


Figure 6: Output screen 3

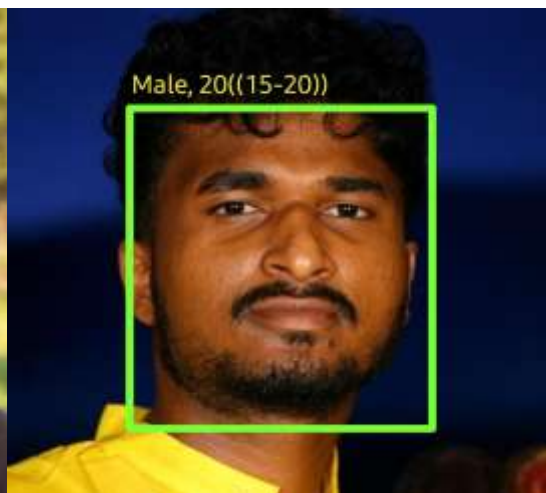


Figure 7: Output screen 4

V. CONCLUSION

This project demonstrates a practical implementation of real-time face detection and age prediction using the OpenCV deep learning module with a Bottle web application. The solution takes advantage of the previously trained models, including files:- Cafe Model

The application provides:

- Webcam integration: live video feed with age and gender prediction.
- PHOTO LOAD: Allows users to load an image for detection.
- Visual comments: shows limited boxes and predictions on the detected faces.
- Flask Backend: a web server that manages video transmission and file loads without problems.

When combining OPENCV DNN, Caffe, and Flask models, this project effectively carries deep learning capabilities to an easy-to-use web interface. This makes it an excellent workforce for computer vision applications in real time and educational purposes in learning how to implement automatic learning models in web applications.

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